

A Surface-Content Decomposition of AI-Generated Text Detection Benchmarks: Evidence from Five Models on DAIGT V2 and HC3

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Abstract—A detector that scores 99% on a machine-generated-text benchmark may have modelled how machine language differs from human language, or how one corpus was punctuated and encoded, and the headline number does not distinguish them. We propose a decomposition that measures the difference. A surface-only model reads 47 orthographic features and never sees a word, a content-only model sees text with punctuation, casing and non-ASCII stripped, and a full model is a fine-tuned transformer on raw text. Across DAIGT V2 and HC3, five model families and eight configurations, the two corpora separate. On HC3 the surface and content arms are indistinguishable under a paired test, 3.20% against 3.26% error, so orthography alone matches content alone. On DAIGT V2 content is stronger by a factor of 7.9. Both arms can read document length, a known confound on both corpora, and closing that channel widens the DAIGT V2 gap to 10.6 times and turns HC3’s parity into an advantage for orthography. Three further controls change what the results mean. BERT emits identical token identifiers for strings differing only by the whitespace cue that dominates discussion of HC3, so it cannot represent that cue and still reaches 0.9916 F1, making the cue sufficient in isolation but unnecessary in practice. A pipeline control confirms that null is not a cleaning step which failed to run. A label-free control shows these models call random character strings human in 100% of cases at 0.98 confidence, so an adversarial collapse to 0.3644 F1 is not a targeted vulnerability. Surface form is informative on both corpora, and we claim only that it reaches parity with content on one.

Index Terms—AI-generated text detection, benchmark evaluation, surface features, shortcut learning, tokenisation, ROC analysis

I. INTRODUCTION

Detectors of machine-generated text report accuracy at or above 97% on the benchmarks the field uses most, and frequently above 99% [1], [2], [3], [4]. Read directly, those numbers say the task is close to solved on the domains these corpora represent. Read carefully, they say something weaker, because a benchmark score measures separability of a particular corpus and not the capability a reader infers from it.

The difference matters because separability has more than one source. A detector can reach a high score by modelling how machine-generated language differs from human language, or by modelling how one corpus was assembled, punctuated and encoded. Both produce the same headline figure. Individual instances of the second have been documented, including a whitespace convention in HC3 [5], a length confound in the same corpus [6], punctuation inconsistencies

requiring standardisation [7], and prompt-specific collection shortcuts [8]. What is missing is a way to ask how much, in total, a given benchmark’s separability owes to surface form rather than content, and to compare benchmarks on that axis.

This paper proposes that measurement and applies it. We compare three arms on identical splits. A *surface-only* model reads 47 orthographic features and never sees a word. A *content-only* model sees text after punctuation, casing and non-ASCII characters have been stripped away. A *full* model is a fine-tuned transformer on raw text. The gap between the first two bounds how much of the benchmark a detector could pass without reading language at all.

Applied to DAIGT V2 and HC3 across five model families, the answer differs sharply between the two. On HC3 the surface-only and content-only arms are indistinguishable, 3.20% against 3.26% test error. On DAIGT V2 content is stronger by a factor of 7.9 in error, 0.99% against 7.86%. The same contrast appears in the model comparison. The best classical configuration on DAIGT V2 falls 0.0007 F1 short of the best transformer, a gap a paired test on the two models’ own predictions cannot separate from zero, and falls short by an error-rate factor of 16 on HC3. That tie is a statement about text budget rather than architecture. The classical model reads the whole essay and the transformer reads its first 128 tokens, and held to one budget the transformer is ahead on both corpora.

We also report two negative results, because both correct claims that this analysis originally made and that the literature invites. The whitespace cue that dominates discussion of HC3 is sufficient in isolation yet unnecessary in practice, since BERT’s tokeniser provably cannot represent it and BERT still reaches 0.9916 F1. And a character-level perturbation that drives DeBERTa from 0.9972 to 0.3644 F1 looks like a targeted vulnerability but is not one. The same model labels uniform random strings as human in 100% of cases at 0.98 confidence. We report that requirement rather than the vulnerability.

Our contributions are the decomposition itself, as a reusable measurement over any human-versus-machine corpus, together with the length control that keeps its two arms from sharing a channel. A second contribution is its application to two widely used benchmarks, with a complete eight-configuration model comparison and per-model ROC, area and confusion analysis. A third is the set of three controls, on tokenisation, on pipeline

execution and on label-free inputs, each of which overturns a conclusion that the uncontrolled version of this study had reached.

II. RELATED WORK

Detection on these benchmarks. Guo et al. [1] introduce HC3 and report a RoBERTa detector at 99.82 F1, establishing both the corpus and the accuracy regime the field has worked in since. Su et al. [2] show that regime is narrower than it appears, since semantic-invariant tasks drop accuracy to roughly 91.7%. Zhou [3] reaches competitive DAIGT V2 accuracy with classical models over character n-grams and no transformer, consistent with our finding but reported without the surface-content separation that would explain it, and without matching the text budget the transformers were given. Kubrusly et al. [9] likewise report strong bag-of-words results on HC3 without isolating what the features encode. Ansary [4] and Lamsiyah et al. [10] report near-ceiling DAIGT-family results, both in-distribution only. Annepaka et al. [11] mix linguistic features with a transformer for 99.45% on HC3, combining the two sources our decomposition separates.

Documented artefacts. Tian et al. [5] identify the HC3 whitespace convention, observe that a detector could score well by exploiting it, and release a cleaning kit. Their contribution is the artefact and the remedy. They do not quantify how much of the corpus’s total separability surface form carries, and they do not test whether a given detector can represent the cue at all, which is the question our tokenisation control answers and which turns out to change the conclusion for BERT. Baidya et al. [6] document a length confound in HC3 and control it by length-matching, again one artefact in isolation, and it is the confound our own arms turn out to share, and Section IV-C closes it. Ardeshirifar [7] standardises punctuation and contractions before comparing handcrafted features to deep models, establishing the practice as necessary without measuring what it removes. Park et al. [8] show HC3-trained detectors rely on prompt-specific collection shortcuts. Where these identify individual cues, we measure the aggregate, which is what a practitioner comparing two benchmarks needs.

Robustness and its controls. Antoun et al. [12] attack HC3-trained detectors with misspellings and homoglyphs and report a fall from 99.88 F1 to 33.57% accuracy on a hand-built adversarial set. Huang et al. [13] report comparable degradation under systematic character perturbation. Neither reports a label-free control, and our results suggest such collapses cannot be separated from a generic degenerate response to unreadable input without one, since our own models label random character strings as human with near-maximal confidence. This is a methodological gap rather than a dispute about their measurements. Lai et al. [14] show ensembling preserves in-distribution accuracy while still degrading under shift, so the fragility is not single-model variance. Zhang et al. [15] reframe detection as a two-sample test, sidestepping single-document shortcuts at the cost of needing a population. Borile and Abrate [16] ablate roughly twenty feed-forward neurons

for up to 6.9 points of out-of-distribution gain, evidence that some in-distribution performance rests on localised corpus-specific features.

Cross-corpus evaluation. Alikhanov et al. [17] merge HC3 and DAIGT V2 into a single 124,195-sample corpus and evaluate under a topic-held-out split, reporting 82.87% to 88.86% accuracy against the near-ceiling numbers usual in-distribution. Their design asks whether one model can serve both corpora. Ours asks what each corpus is separable by, which is why the two results sit alongside rather than replacing one another. Lu et al. [18] show detectors can be evaded by prompting alone, a threat model orthogonal to the corpus-property question studied here.

III. METHOD

A. Datasets

DAIGT V2 [19] contains 44,868 argumentative student essays, 27,371 human-written from the PERSUADE corpus and 17,497 machine-generated by a mixture of 2023-era systems including GPT-3.5, Llama-2, Mistral-7B and Falcon-180B, and was assembled for a public detection competition [20]. The human side derives from the PERSUADE corpus of student argumentative writing [21]. HC3 [1] contains question-answer pairs from five English sources, contrasting human answers with a single generator, GPT-3.5-Turbo-0301, over 85,449 rows [22]. Both were class-balanced for training, giving 34,994 DAIGT V2 rows and 53,806 HC3 rows, and partitioned once into fixed 72/8/20 train, validation and test splits used unchanged by every model reported here, 25,196/2,800/6,998 rows for DAIGT V2 and 38,785/4,289/10,732 for HC3. The partition is duplicate-group aware, so no two documents with identical normalised content are placed on opposite sides of a boundary. Training partitions are balanced to within 0.002 of parity, which matters for interpreting the degenerate-prediction control in Section III-D.

A duplicate-detection pass over the unbalanced corpora finds DAIGT V2 essentially free of internal duplication, six rows in six groups, 0.01%, with no text under both labels. HC3 carries 6,118 duplicate rows in 5,986 groups, 7.16%, plus 476 rows labelled machine-generated that are collection artefacts, typically stored API failure messages. That duplication is why the split is group-aware, and the choice is worth more than it sounds. No HC3 test document recurs in training or validation under the group-aware partition, whereas under a plain stratified split of the same balanced sample 570 of 10,762 do, 5.30%. DAIGT V2 leaks nothing either way. No number in this paper is inflated by memorised duplicates.

B. Models and the combinations evaluated

Five model families are evaluated. Three are classical, Naive Bayes, logistic regression and a linear support vector machine, each run under bag-of-words and TF-IDF for six classical configurations per dataset. Earlier work at full scale reported only the winning representation per model, so we report the complete grid. Two are transformers, `bert-base-uncased` and `microsoft/deberta-v3-base`, fine-tuned over a

grid of learning rate $\{2e-5, 3e-5\}$, batch size $\{16, 32\}$ and weight decay $\{0.01, 0.1\}$, sixteen configurations per dataset, with the operating point selected on validation F1.

Classical models receive text through the normalisation pipeline described below. Transformers receive raw text truncated to 128 tokens, a constraint inherited from the study this corpus was built for and quantified in Section V.

C. The surface-content decomposition

The central instrument of this paper separates how much of a benchmark’s human-versus-machine separability is available from surface form alone and how much from content alone. Three arms are compared on identical splits.

The *surface-only* arm uses 47 orthographic features and never reads word identity. These cover punctuation character rates, whitespace behaviour including the count of spaces preceding punctuation, casing statistics, document and sentence length, non-ASCII and emoji rates, and digit rates. Because no lexical item enters the feature vector, this arm cannot access content by construction.

The *content-only* arm applies the same surface normalisation the classical pipeline begins with, lowercasing and replacing every character outside $[a-z\s]$ with a space, then vectorises to bag-of-words. It applies no stopword removal and no lemmatisation, so it is an unfiltered content model and not a reduced one. Punctuation, casing and non-ASCII characters cannot survive the transformation.

One channel survives into both arms. The surface arm reads character, word and sentence counts directly, and raw bag-of-words rows sum to document length. Length is a documented confound on both corpora, so we close it rather than assume it away, with a 42-feature surface arm stripped of every document-size feature and a content arm whose rows are normalised to unit total and rescaled so regularisation bites equally hard in both. Section IV-C reports what changes.

The *full* arm is the fine-tuned transformer on raw text. Surface-only and content-only share one classifier family, logistic regression, so they are directly comparable to each other. The transformer differs in family and is reported as a reference point rather than as a matched arm, and we draw no conclusion from a transformer-to-classical difference alone.

D. Statistics and confound checks

Every reported figure comes from a committed script writing JSON, and every transformer baseline is read from the deployed checkpoint’s own record, never from the maximum over its hyperparameter grid. An earlier version of this analysis compared a grid maximum against a single fixed configuration and reported differences that did not exist, which is why baselines are now checkpoint-matched throughout.

We report error rate alongside F1 because at this accuracy ceiling F1 differences compress, and macro alongside weighted F1 so that no result can be produced by class skew. Seed spread over three seeds is reported as a range and only ever for the cell it was measured on, never borrowed across datasets or model families, since fine-tuning outcomes

TABLE I

ALL MODEL CONFIGURATIONS, BOTH DATASETS. TRANSFORMER ROWS ARE THE DEPLOYED CHECKPOINTS, NEVER A GRID MAXIMUM, AND CLASSICAL ROWS ARE REFITS ON THE SAME SPLIT. ERROR RATE IS GIVEN BESIDE F1 BECAUSE F1 COMPRESSES AT THIS CEILING. BOLD MARKS THE BEST CLASSICAL AND THE BEST OVERALL ENTRY PER DATASET COLUMN. MACRO F1 EQUALS WEIGHTED F1 TO FOUR DECIMALS IN EVERY ROW, SO NO RESULT IS PRODUCED BY CLASS SKEW.

Model	Rep.	DAIGT V2		HC3	
		F1	err %	F1	err %
Naive Bayes	BoW	0.9591	4.09	0.8713	12.87
Naive Bayes	TF-IDF	0.9577	4.23	0.8670	13.28
Logistic Reg.	BoW	0.9893	1.07	0.9551	4.49
Logistic Reg.	TF-IDF	0.9857	1.43	0.9365	6.35
SVM	BoW	0.9870	1.30	0.9475	5.25
SVM	TF-IDF	0.9910	0.90	0.9449	5.51
BERT	raw	0.9916	0.84	0.9916	0.84
DeBERTa	raw	0.9917	0.83	0.9972	0.28

vary materially with initialisation and data order [23], [24]. It is an order-of-magnitude indication, not a test statistic, so no comparison in this paper rests on it. Where two models are compared on the same test set we use paired evidence instead, exact-binomial McNemar on the discordant predictions together with a paired bootstrap 95% interval on the error difference over 10,000 resamples. Because a bootstrap resamples one test set and says nothing about how a corpus was partitioned, the decomposition is also repeated over five group-aware partitions, varying only the split seed. We do not run parametric tests at $n \leq 30$.

Three controls guard the specific claims made below. First, a *pipeline-fires* control. Our whitespace cleaning produces bit-identical predictions for BERT on HC3, which alone would be indistinguishable from cleaning that silently failed to run. The same pipeline applied to DAIGT V2 does change BERT’s predictions, since emoji removal is visible to its tokeniser, establishing that the pipeline fires and that the null is specific to whitespace. Second, a *tokenisation* control, comparing token identifiers for minimally different strings to establish what each tokeniser can represent. Third, a *label-free* control, scoring 400 inputs per condition that carry no human-versus-machine information at all, including uniform random character strings, token-shuffled text, punctuation-only strings and the empty string. Without this last control a confident prediction on corrupted input is indistinguishable from a learned response to it.

IV. RESULTS

A. A classical model matches a transformer on DAIGT V2 but not on HC3

Table I reports every configuration on both datasets and Fig. 1 the corresponding ROC curves and areas. The two benchmarks differ in a way no single headline number reveals. On DAIGT V2 the best classical configuration, an SVM over TF-IDF at 0.9910 F1, sits 0.0007 below the best transformer at 0.9917. On the same test documents the two disagree on 99 cases, 47 to 52, so McNemar returns $p = 0.69$ and the paired bootstrap puts the error difference at +0.07 percentage

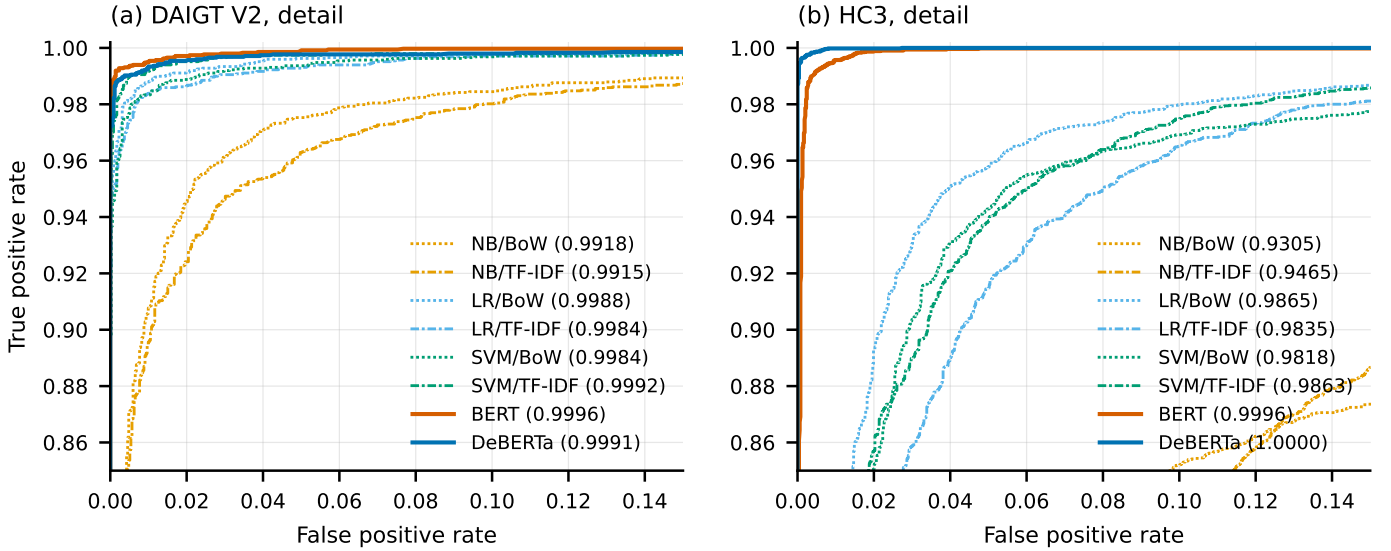


Fig. 1. ROC curves for all eight model configurations, restricted to the low-false-positive region where the models actually differ. Areas under the full curve are given in the legend. On DAIGT V2 (a) every configuration converges toward the corner and the classical models are close behind the transformers. On HC3 (b) the two transformers separate sharply from all six classical configurations. Colours are colourblind-safe and each series carries a distinct dash pattern, so the ordering survives greyscale reproduction.

points with a 95% interval of $[-0.21, +0.34]$ that contains zero. On HC3 the same comparison is 0.9551 against 0.9972, with 484 discordant cases split 16 to 468, $p < 10^{-6}$, and an error difference of +4.21 points with interval $[+3.82, +4.61]$.

That DAIGT V2 tie is not a matched comparison, and reading it as an architecture result would be wrong. The classical models receive the whole document while the transformers receive 128 tokens, which is 33.5% of a mean DAIGT V2 essay against 74.6% of a mean HC3 answer. Refitting the classical grid on the exact span each checkpoint’s own tokenizer kept removes the asymmetry. The best DAIGT V2 configuration then rises from 0.90% to 2.10% error against BERT’s 0.84%, a difference of 1.26 points, interval $[+0.90, +1.63]$, $p < 10^{-6}$, and the DeBERTa window returns the same 1.26. On HC3 the gap only widens, 4.49% to 5.66%. Held to one text budget the transformers lead on both corpora. What the unmatched comparison shows is narrower and still worth stating, that a bag-of-words model reading a whole essay matches a transformer reading its opening.

Ranking by area does not reproduce ranking by F1. On DAIGT V2, BERT attains the higher area, 0.9996 against 0.9991, while DeBERTa attains the higher F1. Neither ordering is real, since the two disagree on 75 documents, 37 to 38, $p = 1.00$. On HC3 the pair separates cleanly at $p < 10^{-6}$.

B. On HC3, orthography alone matches content alone

Table II gives the decomposition. On HC3 the two arms are statistically indistinguishable, 3.20% error from orthography against 3.26% from content. They disagree on 625 documents, split 316 to 309, so McNemar returns $p = 0.81$ and the paired bootstrap puts the difference at -0.07 percentage points with a 95% interval of $[-0.52, +0.40]$. Forty-seven features that never read a word do as well as a bag-of-words model over

TABLE II
SURFACE-CONTENT DECOMPOSITION. SURFACE-ONLY USES 47 ORTHOGRAPHIC FEATURES AND NEVER READS A WORD. CONTENT-ONLY IS UNFILTERED BAG-OF-WORDS AFTER PUNCTUATION, CASING AND NON-ASCII ARE STRIPPED. THE LOWER BLOCK CLOSES THE DOCUMENT-LENGTH CHANNEL THAT BOTH PRIMARY ARMS CAN OTHERWISE READ. BOLD MARKS THE BETTER ARM IN EACH BLOCK AND DATASET COLUMN. THE TRANSFORMER IS A REFERENCE POINT, NOT A MATCHED ARM.

Arm	DAIGT V2		HC3	
	F1	err %	F1	err %
surface-only	0.9214	7.86	0.9680	3.20
content-only	0.9901	0.99	0.9674	3.26
full transformer	0.9917	0.83	0.9972	0.28
<i>length channel closed</i>				
length-only (3 feats)	0.7540	24.59	0.8475	15.23
surface-only, no length	0.9007	9.93	0.9662	3.37
content-only, length-norm.	0.9906	0.94	0.9371	6.28

the whole vocabulary. On DAIGT V2 the same two arms separate by a factor of 7.9 in error, 7.86% against 0.99%, and there the separation is unambiguous, with 569 discordant documents split 44 to 525, $p < 10^{-6}$, difference +6.87 points with interval $[+6.23, +7.52]$.

Surface form is informative on both benchmarks, since DAIGT V2’s surface-only arm reaches 0.9214 F1, far above chance. The finding is not that one corpus is clean and the other is not, but that only on HC3 does surface rise to parity with content.

One partition could produce this by chance, so we repeat the decomposition over five group-aware partitions of the same sample, varying only the split seed. Content leads on DAIGT V2 on all five by 6.00 to 6.87 points, $p < 10^{-6}$ throughout. On HC3 the difference is significant on none of the five, p between 0.23 and 0.81, and changes sign between them, which is what

a null looks like and an underpowered real difference does not.

The content-only arm reaches 0.9901 on DAIGT V2 against the transformer’s 0.9917, within 0.16 points of error, under the same unmatched text budget as above. On HC3 the transformer improves on either arm by roughly 0.03 F1, consistent with combining information neither arm holds alone.

C. Neither result is an artefact of document length

Both arms can read document length, so the comparison above could be a length comparison in disguise. Length alone reaches 24.59% error on DAIGT V2 and 15.23% on HC3, well short of either arm but far from nothing. Closing the channel on both sides sharpens rather than weakens the two conclusions. On DAIGT V2 the length-free content arm loses nothing, 0.94% against 0.99% at $p = 0.71$, while the length-free surface arm rises to 9.93%, widening content’s advantage from 7.9 to 10.6 times. On HC3 the surface arm barely moves, 3.20% to 3.37%, while the content arm rises to 6.28%, so orthography no longer merely matches content but beats it by 2.91 points, interval $[+2.37, +3.44]$, $p < 10^{-6}$. Removing length costs the content arm eighteen times what it costs the surface arm, the opposite of a length-driven surface result, and orthography leads HC3 on all five partitions by 2.67 to 2.91 points. One caution on the instrument. Normalising bag-of-words rows to unit total shrinks every feature by roughly the mean document length, so at fixed regularisation the penalty becomes far heavier and the arm collapses to 9.69% and 14.25%. That is an artefact of scale, not length, and the figures above use rows rescaled to the raw-count arm’s magnitude.

D. A model that cannot represent the cue reaches 0.9916 regardless

The most-discussed HC3 artefact is a space preceding punctuation in human answers [5], present in our test split at 10.745 occurrences per human document against 0.013 per machine one, and in 88.7% of human documents against 0.3%. Tokenisation makes the natural assumption that detectors exploit it testable rather than rhetorical.

BERT’s WordPiece tokeniser splits on punctuation irrespective of adjacent whitespace. For "the answer is simple ." and "the answer is simple." it emits identical identifier sequences, in three of three pairs tested, while DeBERTa’s SentencePiece encodes leading whitespace and distinguishes all three. BERT cannot represent the cue at all and still reaches 0.9916 F1 on HC3, so it is sufficient in isolation and unnecessary in practice.

This is also why removing it changes nothing for BERT. Applying whitespace cleaning to HC3 yields predictions bit-identical to the uncleaned run, which taken alone would be indistinguishable from a cleaning step that never executed. The same pipeline applied to DAIGT V2, where it removes emoji that WordPiece does represent, does change BERT’s predictions. The pipeline fires, and the null is specific to whitespace.

We do not attribute the BERT-to-DeBERTa difference on HC3 to this cue. The two models differ in tokeniser, architecture, pretraining corpus and parameter count, and this experiment separates none of those.

E. The adversarial collapse is not distinguishable from a degenerate response

Perturbing HC3 test text with character-level typos at 10% of alphabetic characters drives DeBERTa from 0.9972 to 0.3644 weighted F1. The confusion matrix is one-directional, $[[5376, 1], [5207, 148]]$, with human documents still classified perfectly and machine documents reassigned to human. Read alone this looks like a targeted vulnerability. It is not what the control shows.

On inputs carrying no human-versus-machine information, the same model assigns *human* to uniform random character strings in 100% of cases at mean confidence 0.98, and to token-shuffled and punctuation-only text likewise. All four models label the latter two as human between 94.8% and 100% of the time, so the behaviour is specific to neither HC3 nor the attack.

Three candidate mechanisms were tested and none survives. Majority-prior collapse is excluded by the 0.002 training balance. Cue injection is excluded because the perturbation does not create the whitespace artefact, moving machine-document counts only from 0.000 to 0.005. A learned disfluency-implies-human association is unsupported, since subword fertility runs the wrong way, 1.1192 for HC3 human text against 1.1980 for machine.

We therefore decline to read our own perturbation numbers as robustness measurements. The experiment establishes a requirement on future work, not a property of these benchmarks.

V. DISCUSSION AND CONCLUSION

The decomposition separates two benchmarks that a headline accuracy figure does not. On HC3, orthography alone matches content alone, 3.20% against 3.26% test error. A paired test cannot distinguish that 0.07-point difference from zero. A fine-tuned DeBERTa improves on either arm by roughly 0.03 F1. On DAIGT V2, content is stronger than orthography by a factor of 7.9 in error. The best classical model there is within 0.0007 F1 of the best transformer, a gap the same test also cannot separate from zero, though only because the two read different amounts of text. A practitioner choosing an evaluation corpus, or a reviewer reading a reported score, cannot recover that distinction from the score itself.

What the measurement does not license is a claim that either corpus is clean. DAIGT V2’s surface-only arm reaches 0.9214 F1, far above chance, so surface form is substantially informative on both. The finding is comparative, that surface reaches parity with content on one corpus and not the other, and it is bounded by the arms we built. A richer surface featurisation would raise both surface numbers, and a stronger content model would raise both content numbers, so the arms bound rather than partition the separability.

The two arms are also not disjoint as built, since document length reaches both. Closing that channel weakens neither conclusion. DAIGT V2’s content advantage widens from 7.9 to 10.6 times, and HC3’s parity becomes a 2.91-point advantage for orthography, since removing length costs the content arm eighteen times what it costs the surface arm. The control nearly produced a third false conclusion of its own, since normalising bag-of-words rows without restoring their scale collapses the arm through regularisation and reads as a length effect.

Two results here began as the opposite conclusion, and both are worth stating for what they say about method. A cleaning experiment that reports no change is uninterpretable without evidence that the pipeline ran at all. A tokeniser that cannot represent the cue being cleaned makes the null a fact about the model, not about the corpus. A collapse under perturbation is uninterpretable without a label-free control, since a model that calls random strings human at 0.98 confidence will produce the same curve with no vulnerability behind it. We report both requirements rather than the findings they dissolve, and suggest comparable published collapses be read with the second control in mind.

Several limits bound every number above. The transformers see 128 tokens, so the DAIGT V2 transformer results describe classification from an essay’s opening. We measure that rather than assume it, by refitting the classical grid on the same window, and it reverses the classical-versus-transformer reading on DAIGT V2. The cross-corpus comparison remains not like-for-like in the text each model receives. Seed spread is estimated from three seeds, and one such estimate moved by 31% when its runs were repeated. No claim here rests on it, and every comparison uses a paired test instead. Neither corpus transfers to the other, costing 0.0833 to 0.2019 mean F1 over three seeds. HC3 has one generator collected at one time, and both corpora are English. The content arm keeps stopwords and inflections, so it is not a reduced content model. It is still a bag-of-words one, and word order and syntax are outside it, so it bounds content-carried separability from below for that reason rather than through its preprocessing. We evaluate two corpora and five model families, and nothing here establishes that the pattern holds beyond them.

The measurement itself is the portable part. It needs no new annotation and runs in minutes on a laptop. It yields one number per corpus that a headline accuracy cannot express, namely how much of that corpus a model could pass without reading the language.

DATA AND CODE AVAILABILITY

Both datasets, DAIGT V2 and HC3, are public, and code is released for reproduction. The link is withheld for blind review.

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